

Division : Angiospermae

General characteristics

1. Plant body is sporophytic and differentiated into roots, stems, leaves, flowers and fruits.
2. They may be annual, biennial or perennial herbs, shrubs, climbers or trees.
3. Annual- life span is one year e.g. mustard , biennial- two years e.g. radish and perennial- several years e.g. mango
4. They are herbs , short heighted e.g. grass, shrub –middle height e.g. Rose and tree long.
5. They are provided with vascular tissues, xylem and phloem.
6. They have flower as reproductive structure.
7. Male reproductive part is androecium, and female reproductive part is gynoecium.
8. They are heterosporous: male spores are small called microspore or pollen grains, female spores are large called megaspores.

8. Fertilization takes place by the formation of pollen tubes.
9. Double fertilization and triple fusion is the characteristic of angiosperm.
10. Seeds are enclosed inside the fruits.

Seeds may be albuminous or exalbuminous and consist of one or two cotyledons.

Classification

Angiospermae is divided into two classes;

- Class 1. Monocotyledonae
2. Dicotyledonae

Morphology of Angiosperm

Plant morphology is the study of the external characters of both vegetative as well as reproductive parts of the flowering plants.

The vegetative parts of the plant include two major organ system:

Aerial shoot system and underground root system.

The root system

1. It is the descending parts, non-green part of the vascular plants which grows downward into the soil.
2. It is negative phototropic and positive geotropic.
3. The root is not differentiated into nodes and internodes.
4. Usually, the tip of root is protected by root cap. (hydrophytic plants have root pockets).

5. The root bears unicellular root hairs for absorption of water and minerals from the soil.

6. Lateral roots are endogenous in origin, i.e. they arise from the inner tissue called pericycle of primary root.

The root system is of two types

1. Tap root

2. Adventitious root

Tap root

The root which arises from the radicle of embryo of seed and develops into primary root, secondary and tertiary root is tap root or true root.

Tap root system is the characteristics of dicot plants.

Adventitious root

The root that develops from any part of the plant body except the radicle is called the adventitious root.

The adventitious roots are surficial and spread in a larger area.

It forms a cluster of roots, which may originate from the base of the stem or nodes or internodes.

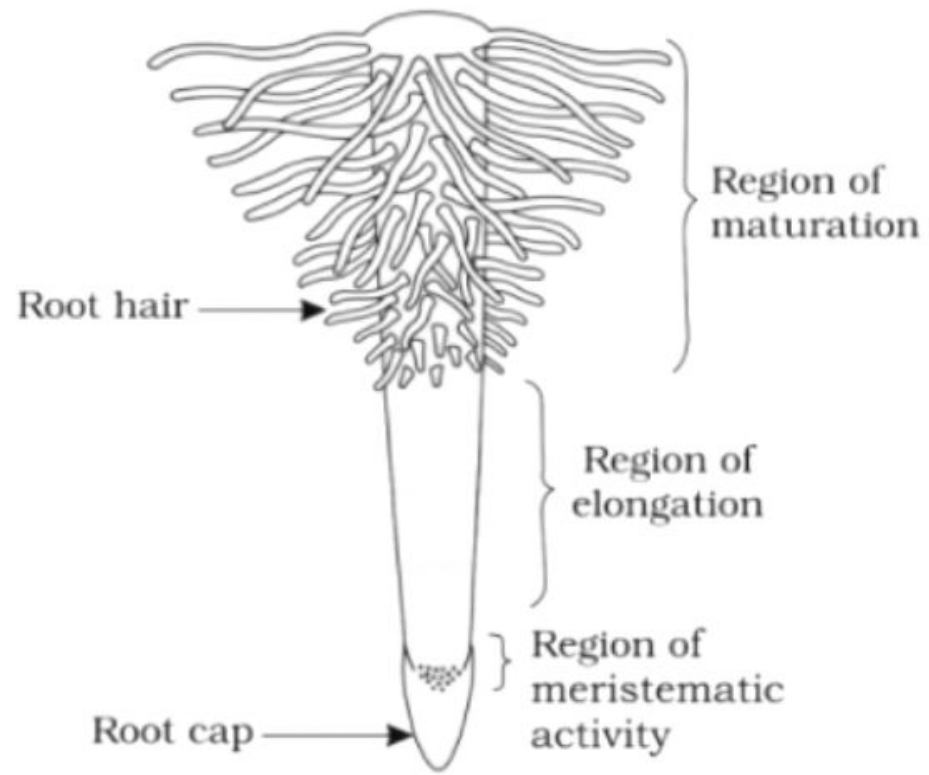
The cluster of adventitious roots having nearly equal size and length are collectively called fibrous root system.

It is commonly present in monocotyledonous plants like wheat, maize etc.

Parts of roots

A typical plant root system has four main regions or zones: the region of root cap, the region of meristem, region of elongation and region of maturation or differentiation.

1. Root cap (calyptra) : It protect the sensitive growing cells; it also provide friction less penetration of root in soil particles.
2. Region of meristem: This region is made up of small, thin-walled cells having dense protoplasm. The cells of this region divide actively to new cells and help in root elongation.
3. The region of elongation: It is located next to the meristematic region. This region consists of elongated cells and increase the length of root. The cells of this region do not divide.
4. Region of maturation: This region is located next to the region of elongation and composed of matured cells . From the epidermal layer of this region root hairs are raised . So, also called as Piliferous region.



The regions of the root-tip

Basic function of root

1. Root system provide fixation to the plants.
2. Root absorb water and minerals from the soil.

Except these primary functions various secondary functions are performed by the root like storage of food, extra support and vital functions. For the secondary functions the roots are modified.

Modifications of Tap root

- A. For storage of food
 - i. Fusiform: The root is swollen in the middle and tapers gradually at both the base and the apex .e.g. *Raphanus sativus*
 - ii. Napiform: It is thick, fleshy and almost globular at base and tapers sharply at the base. The stem is reduced into a disc e.g. turnip.
 - iii. Conical: The primary root is broad at the base and tapers gradually towards the apex. It looks like a cone. The broad apex meets with the reduced disc-shaped stem e.g. carrot.
 - iv. Tuberous: The primary root becomes thick and fleshy, but it has no definite shape. E.g. *Mirabilis*.

B . For Nitrogen Fixation

Nodulated or tuberculated taproots:

In legume plants, secondary roots and sometimes primary roots bear many small or large irregular swellings called root nodules.

These nodules contain nitrogen fixing bacteria (*Rhizobium leguminosorum*). These bacteria help in nitrogen fixation.

e.g. Pea.

C. For respiration

Pneumatophores or respiratory roots:

In some plants growing in marshy soil the branches grow vertically upward in air. They have pores called as Pneumatophores for respiration.

e.g. Rhizophora(mangrove plants).

D. For support

Butress root: In some plants like *Ficus religiosa*(peepal), *Ficus elastica*(rubber), thick woody laterally compressed horizontal roots arise from the base of the main stem for additional support, called as buttress root.

E. For vegetative propagation:

Reproductive root: The tap root of some plants like *Dalbergia sissoo*

Develop adventitious buds for vegetative propagation called reproductive root.

F. For nutrients absorption

Mycorrhizal roots:

In some higher plants the fungi live symbiotically in branches called as mycorrhizal root and fungi are called as mycorrhiza.e.g. *Betula utilis*.

Modification of Adventitious root

- A. For storage of food
 - i. Tuberous root-e.g., sweet potato
 - ii. Fasciculate root e.g., Asparagus, Dioscorea
 - iii. Moniliform or Beaded root: e.g., Bitter gourd
 - iv. Palmate root: e.g., Orchis
 - v. Nodulose root: e.g., *Curcuma amada* (mango ginger)
 - vi. Annulated root: e.g., Ipecac



Tuberous
root



Fasciculated
root



Moniliform
root



Annulated root



Nodulose root



B. For mechanical support

1. Stilt roots: These are short roots which arise from the first few nodes of stem. They penetrate obliquely down into the soil and give support to the plants. e.g., Maize, sugarcane.
2. Prop roots: The roots are pillar-like structure hanging vertically downward from branches and penetrate soil. Gradually the roots become thick and provide mechanical support to the plants . e.g. *Ficus bengalensis* (banyan tree).

C. For vital function

1. Parasitic or haustoria root: They help to absorb nutrients from host plants. Also called as suckers and present in parasitic plants . e.g. *cuscuta*

2. Epiphytic root:

Epiphytes are the plants that grow on branches or on trunks of other plants and roots of them are called epiphytic root. They are also called as hanging roots . Pores called Velamens are present on them which help in absorption of moisture present in air. e.g orchids.

3. Respiratory roots: Mangroove.

4. Floating roots : The adventitious roots absorb air and make the plants floating. Present in hydrophytes. e.g. Pisti , Eichornia.

5. Assimilatory roots or photosynthetic roots: The adventitious root of some hydrophytes help in respiration. E.g. Trapa, Tinospora.

6. Reproductive root : The adventitious root of some plants develop bud which give rise o leaf and shoot .The roots help in vegetative propagation. e.g sweet potato.

7. Root thorns: The adventitious roots developing from stem are modified into thorns for protection . e.g money plant.

8. Climbing or clinging root: These roots arise from nodes in climbers and help them in climbing on substratum. E.g. betel

9. Foliar root (epiphyllous root): Develop from leaf and also help in vegetative propagation .e.g. bryophyllum, begonia.

Shoot system

The stem along with the branches, leaves, flowers and other appendages is known as shoot system.

The stem is the aerial part of the main axis of plants, developed from the plumule of an embryo or the germinating seeds.

The color of young stem is green and become brown and woody as matures.

Characteristics of stem

1. The stem is the ascending part of the main axis of the plant which directly develops from plumule of the embryo.
2. The stem is positive phototropic and negative geotropic.
3. The stem is differentiated into nodes and internodes.
4. The branches and leaves arise from the nodes of the stem.
5. The growing apex bears a terminal bud.

6. The lateral branches are exogenous in origin.

Functions of stem

1. The stem form shoot system.
2. In some cases, stem bears multicellular hairs called as Trichomes.
3. Stem conduct water and food .
4. They can perform photosynthesis and in modified form help in reproduction.
5. It bears leaf, flower and fruits.

Terminologies for stem

Erect – Vertical growing. maize

Prostrate-Horizontal growing. Also called creepers .e.g. Oxalis

Branched – main axis contain branches .e.g. mustard

Unbranched- main axis do not bear branches. e.g.rice

Cylindrical- rounded.e.g. mustard

Angular- stem with ridges.e.g. marigold.

Solid-stem filled with compact cells. E.g. maize

Fistular- hollow ste. E.g.Peas

Herbaceous-Weak, annual herb plant. E.g. Pea.

Woody-Strong and hard stem of shrub and tree. E.g. rose

Glabrous-stem without hair. E.g. Wheat

Hairy- stem with hairs. E.g. sunflower.

Glaucous-smooth and shiny stem.e.g. citrus

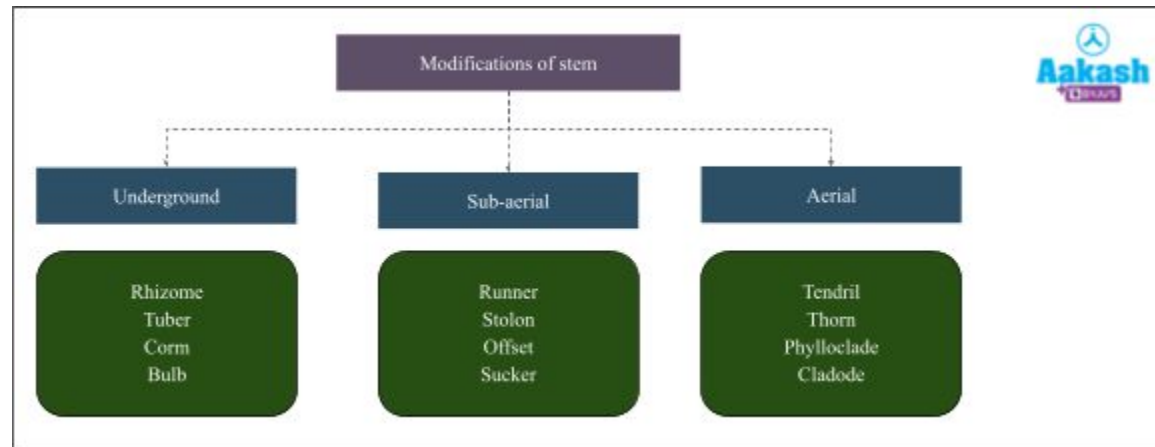
- **Shoot Modifications**

- Shoots modifications can be of three types:

- A. Underground

- B. Sub-aerial

- C. Aerial



- **Underground Modifications**

- These help to store reserve food, propagate vegetatively and also help in perennation.

- **Rhizome**

- Rhizome is a stout, fleshy and horizontal underground stem which stores food. **Nodes** possess axillary buds which are covered by **scaly leaves**. Internodes are prominent. **Adventitious roots** develop at the nodes. Examples: Banana, ginger, turmeric, etc.

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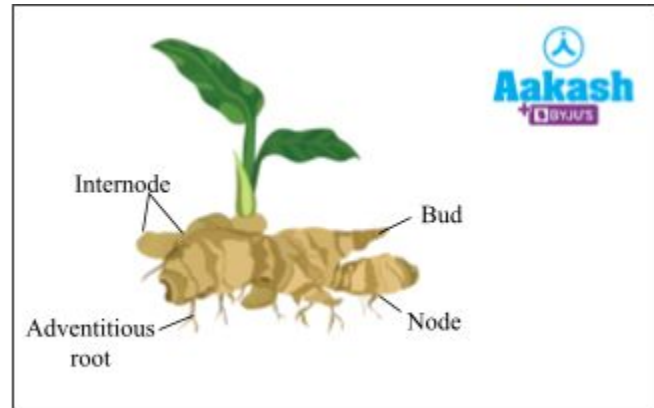


Fig: Ginger - Rhizome

- **Tuber**

- Tubers are swollen ends of special underground branches of stem. These are stouter than rhizomes, with slender internodes and no **adventitious roots**. They bear **axillary buds** covered with scaly leaves in depressions known as **eyes**. Examples: Potato, *Dahlia*.

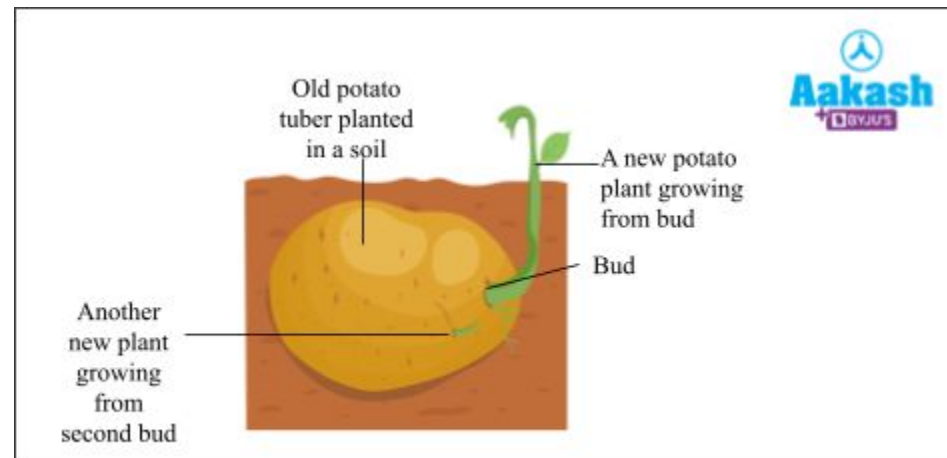


Fig: Potato -Tuber

- **Corm**

- Corm is a short, stout, and usually unbranched stem that grows vertically. It bears many buds in axils of scale leaves. **Buds** grow into aerial shoots and new corms are formed when food gets stored at the base.

Examples: *Amorphophallus* (zaminkand), *Colocasia* (arbi).

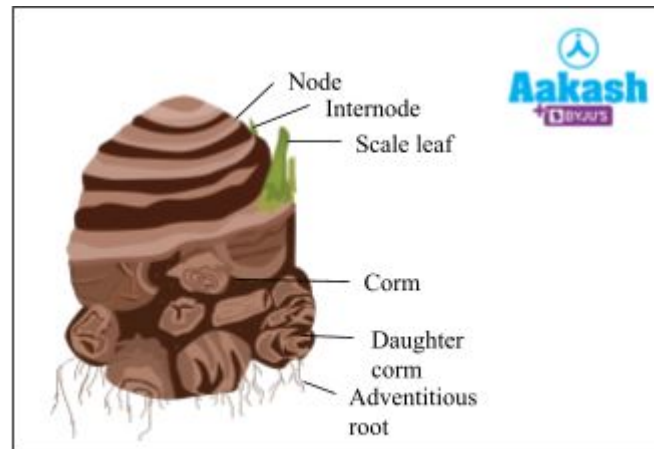


Fig: *Colocasia* - Corm

- **Bulb**

- Bulb is a reduced disc-like stem with adventitious roots present at its base. The stem is covered with overlapping leaves or leaf bases known as **scales** which enclose an apical bud. The scales may be fleshy and **tunicated** or scaly. Examples: Onion, garlic, tulip.

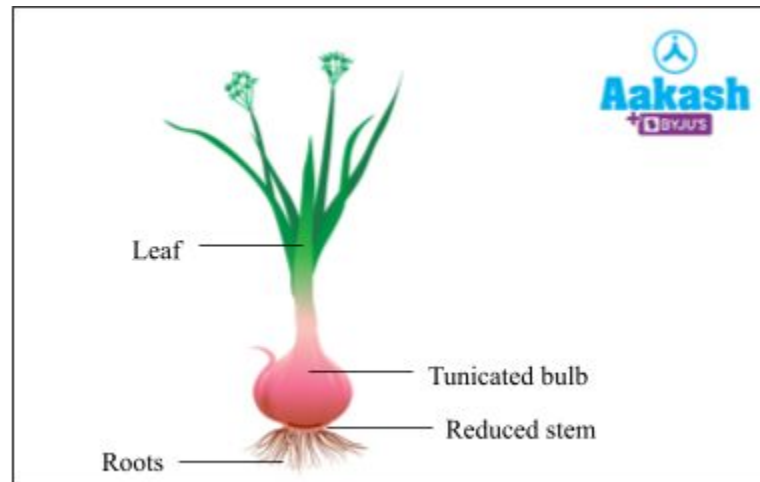
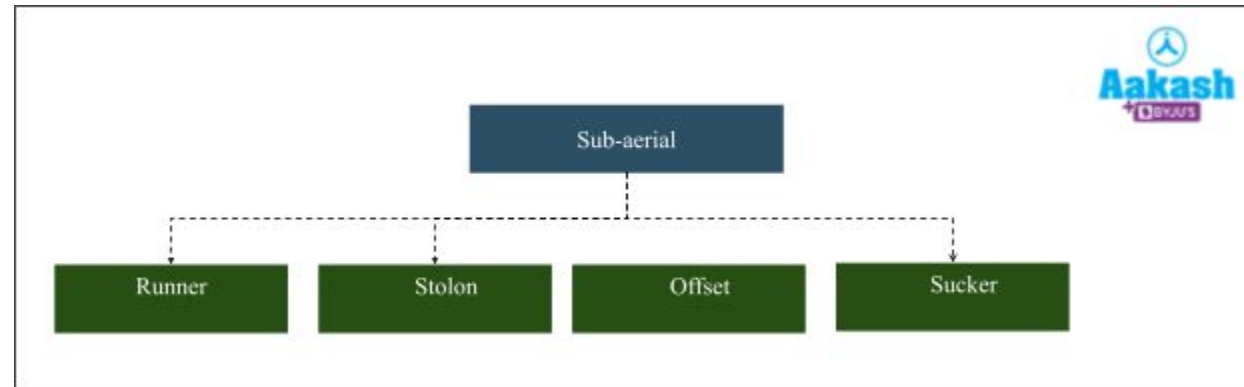


Fig: Bulb of onion

- **Sub-aerial modifications**

- Sub aerial stem modifications are seen in herbaceous plants with thin and weak stems. These mainly help in vegetative propagation.



- **Runner**

- Runners arise from the basal internodes of the aerial branches known as crown and run horizontally over the ground. A runner has long and thin **internodes** and branches creep over the surface of soil. Adventitious roots develop at the **nodes** and aerial branches arise from the axils of **scale leaves** present at the nodes. On getting separated from the parent plant, the daughter plants propagate in a similar manner. Example: Strawberry, grass.

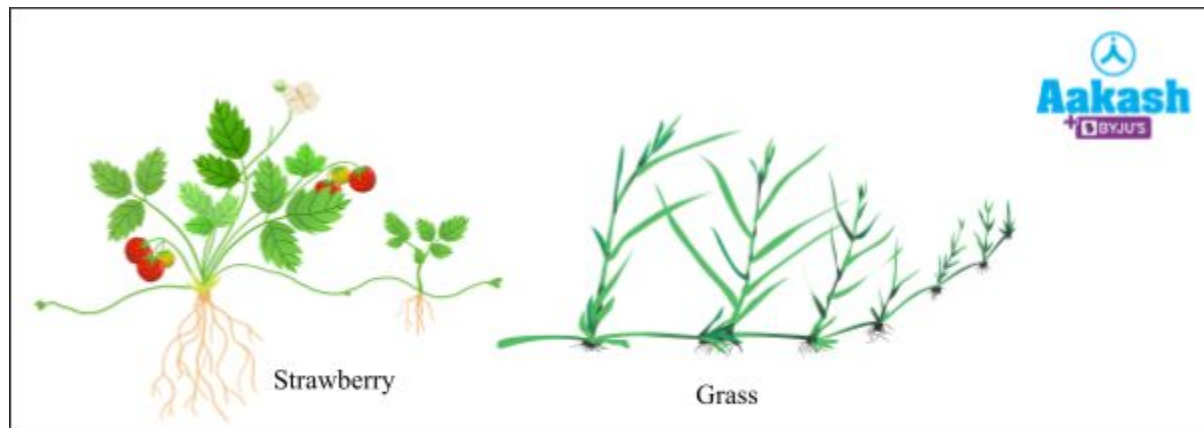


Fig: Runner

- **Stolon**

- Stolon is a weak lateral branch that arises from the base of the main stem. After they grow aurally for some time, they bend downwards to touch the ground, where their terminal bud gives rise to a new shoot and **adventitious roots**. Stolon bear **nodes** and **internodes**. They are like runners except for the fact that they are produced just beneath the surface of the ground. Example: Mint.

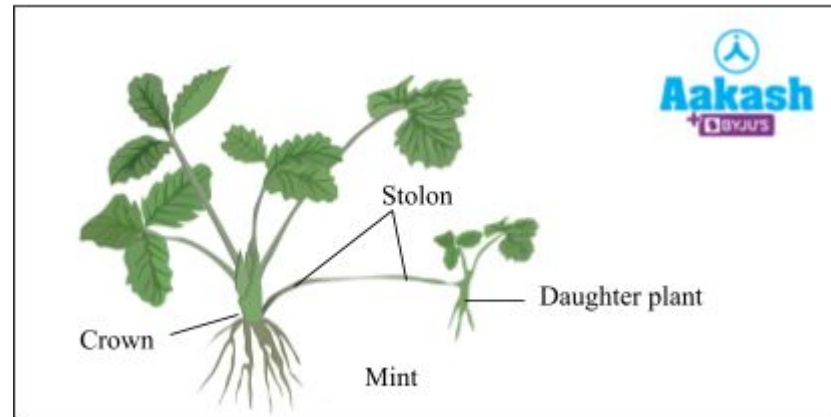


Fig: Stolon

- **Offset**

- Offsets are short, more or less thickened, horizontal branches with short **internodes**. These arise from the **axil** of the lower leaves of the main shoot. At the nodes, the offset produces a tuft of leaves, arranged in a rosette fashion, above and a cluster of **roots** below. Upon detaching from the parent plant, each branch can grow into a new plant. Examples: *Pistia*, *Eichhornia*.

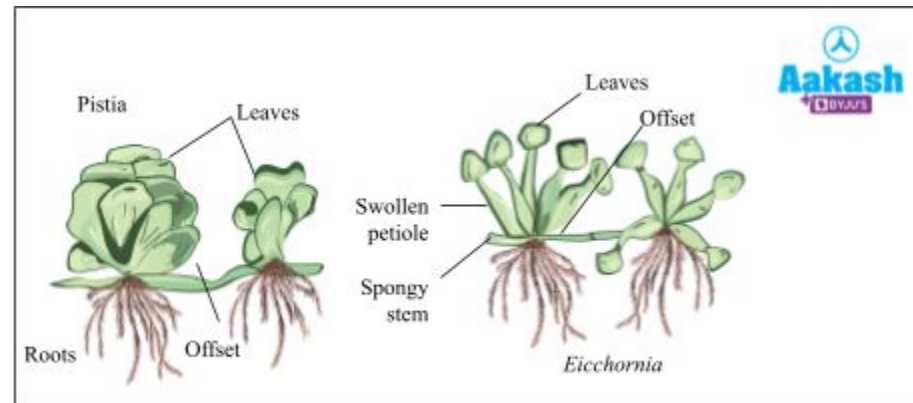


Fig: Offset

- **Sucker**

- The sucker originates as a lateral branch from the underground axillary bud of an aerial shoot. It grows underground obliquely for some distance before emerging upwards. Suckers bear **axillary buds** at the **nodes**, which can give rise to new plants. Examples: *Chrysanthemum*.

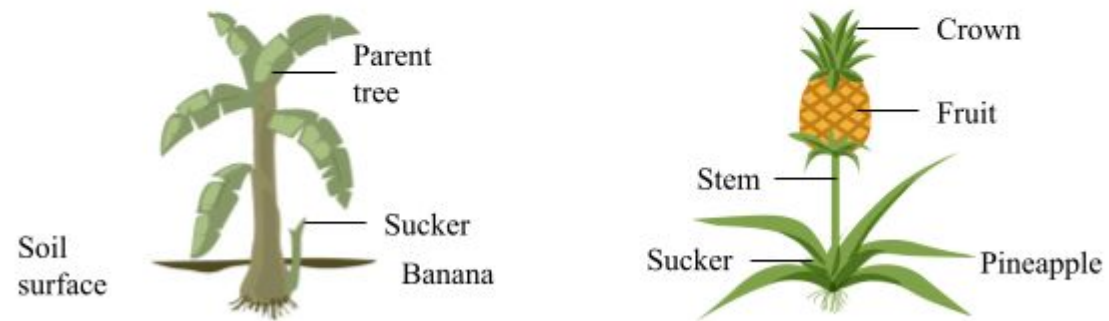


Fig: Sucker

Modification of stem

Based on habit, the stem modifications can be classified into the following three categories:

Underground, subaerial and aerial modification

Underground stem modifications

It is the part of stem which is inside the soil. It resembles like the root but is identified as stem due to the presence of nodes, internodes, scaly leaf, buds and branches.

Based upon the type of growth underground stems are classified into four types. They are

1. Rhizome
2. Corm
3. Tuber
4. Bulb

1. Rhizome

It is stout, horizontal, thick and fleshy underground stem . It has dry scaly leaves for protection of axillary buds , differentiated into nodes and internodes, due to presence of axillary buds it help in vegetative propagation. E.g. ginger, turmeric.

2. Corm

It is vertical growing stem which is stout, swollen, and fleshy. It has scaly leaf at each nodes, it has internodes. The nodes bear buds for vegetative propagation.e.g.colocosia, amorphophyllus.

3. Tuber: They are oval or spherical fleshy underground modification of stem. Here the underground part of stem become branched, and the terminal end of each branch is swollen into tuber with number of eyes with buds which help in vegetative propagation. E.g. potato

4. Bulb

It is short , condensed disc shaped stem which is present at the base of fleshy scale leaves and has axillary bulb which help in vegetative propagation. E.g.Onion

Bulb are of two types:Tunicated and scaly

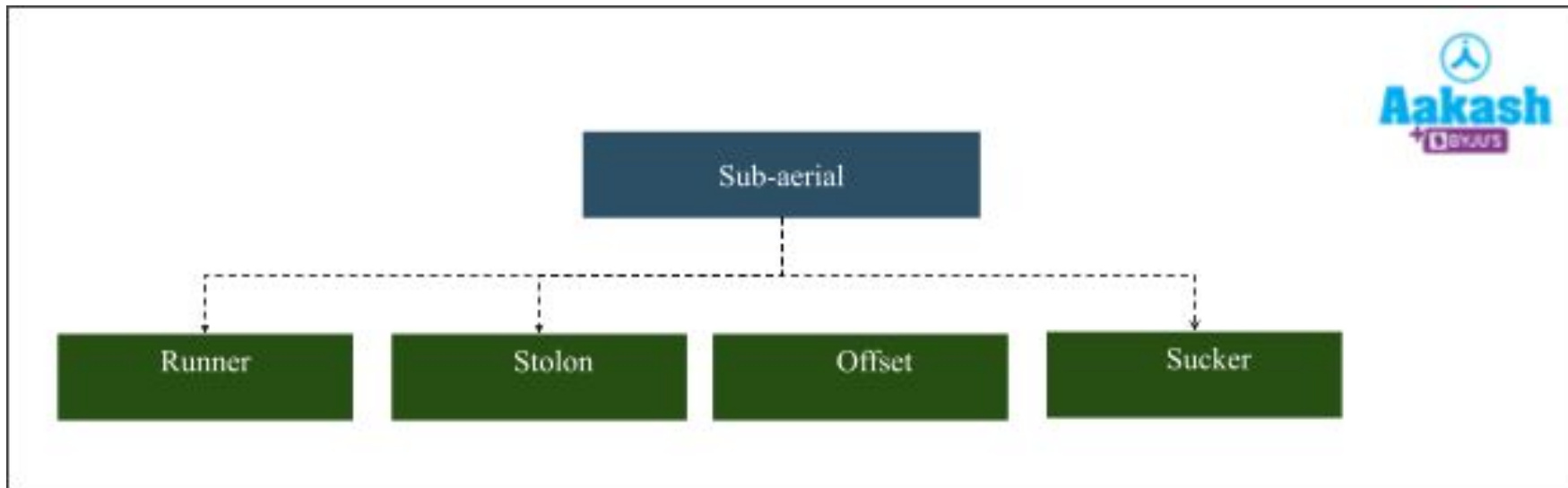
Tunicated bulb is of two types: Simple tunicate ,e.g. onion

Compound tunicated , e.g. garlic

Scaly_Lily

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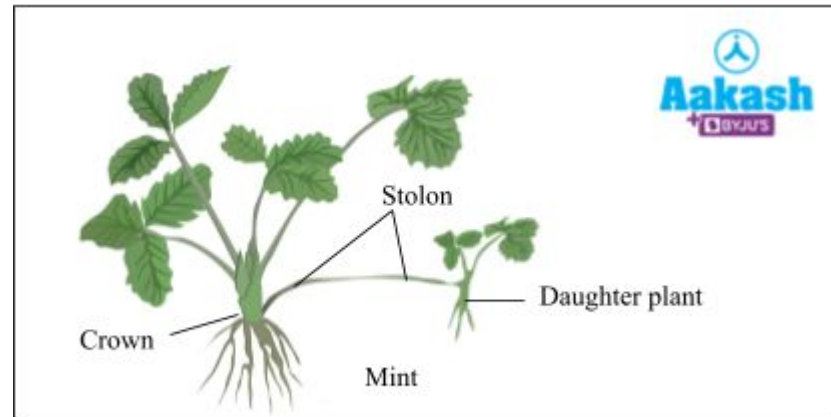


Fig: Stolon

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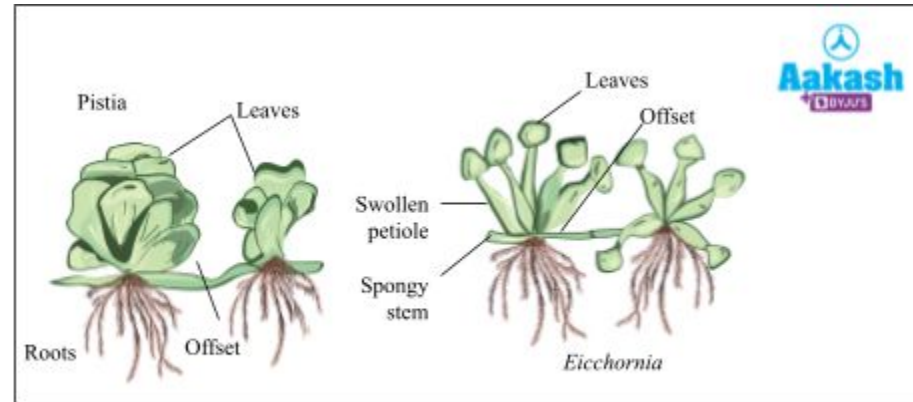


Fig: Offset

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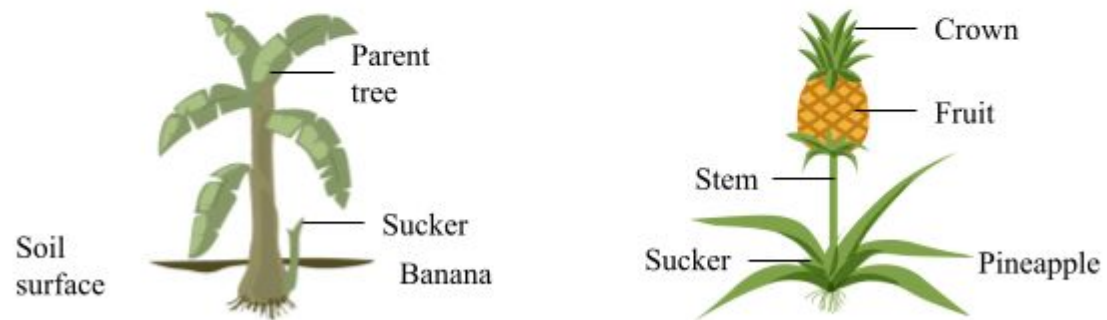
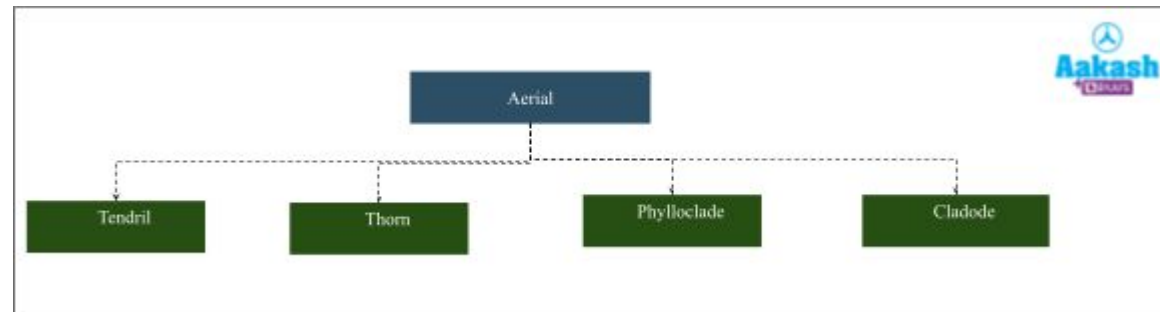


Fig: Sucker

- **Aerial modifications**

- In some plants, the stem above the ground undergoes modifications to perform specialized functions such as providing support, protection and performing photosynthesis.



- **Tendrils**

- These are green thread-like and spirally coiled structures which are sensitive to touch. These help the weak stemmed plants to climb up against a solid support. Examples: Gourds (cucumber, pumpkins, watermelon) and grapevines

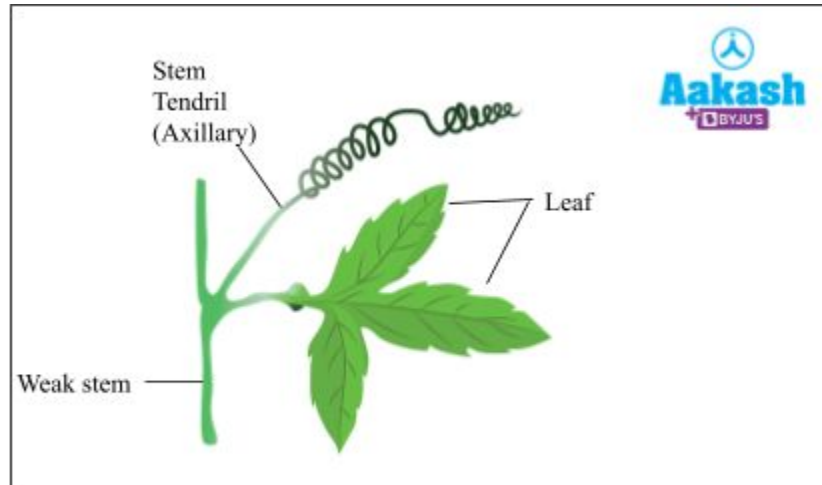


Fig: Stem tendrils

- **Thorns**

- They are hard, pointed structures that develop on the axils of leaves. Thorns are modified **axillary buds**. They provide protection against grazing animals.
Examples: *Citrus*, *Bougainvillea*.



Lemon



Bougainvillea



Fig: Plants with thorns

- **Phylloclade**

- Phylloclade's are flattened or cylindrical green stems of unlimited growth. These are commonly found in **xerophytic** plants in which the leaves are modified to **spines**. Phylloclade can perform **photosynthesis** and can also store water and mucilage. Examples: Thick, fleshy, and succulent in *Opuntia*, cylindrical in *Euphorbia*, *Casuarina*

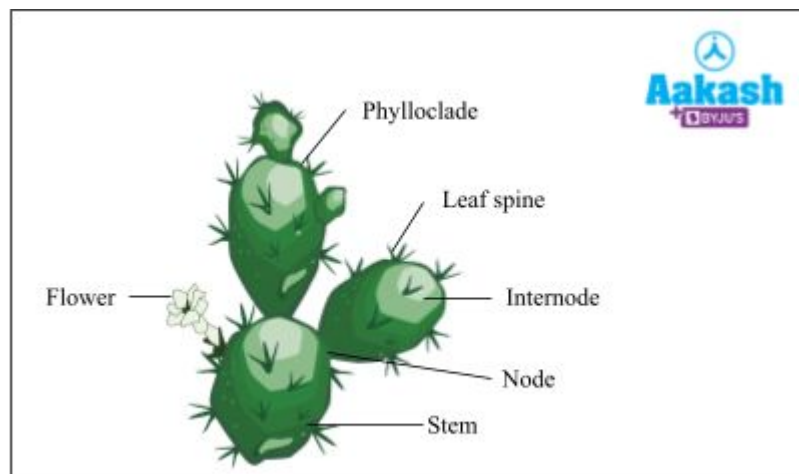


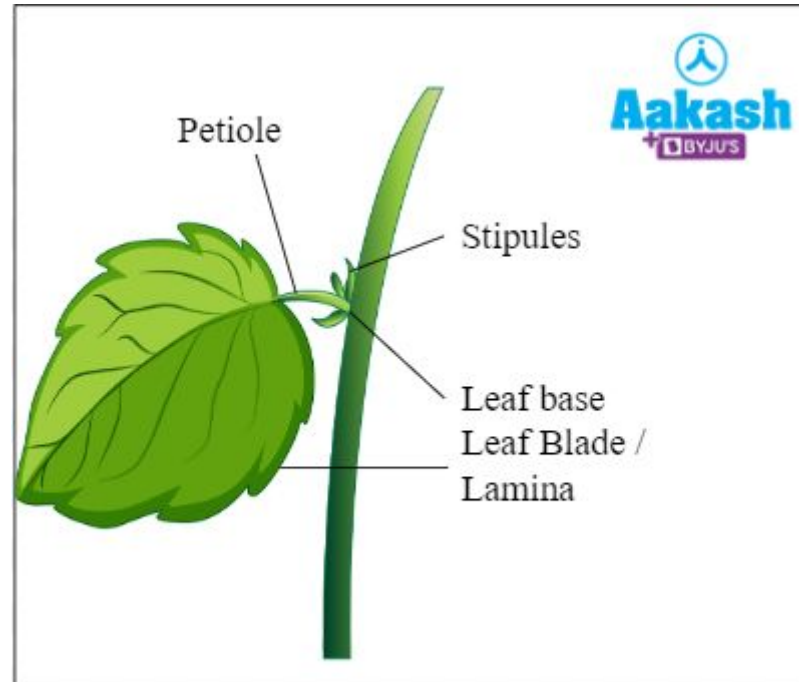
Fig: *Opuntia*

- **Cladodes**

- Cladodes are flattened green stems that are one or two **internodes** long. It develops from the **axil** of a **scaly** leaf. It performs **photosynthesis** as the true leaves are reduced to **spines** or **scales**. Example - *Asparagus*

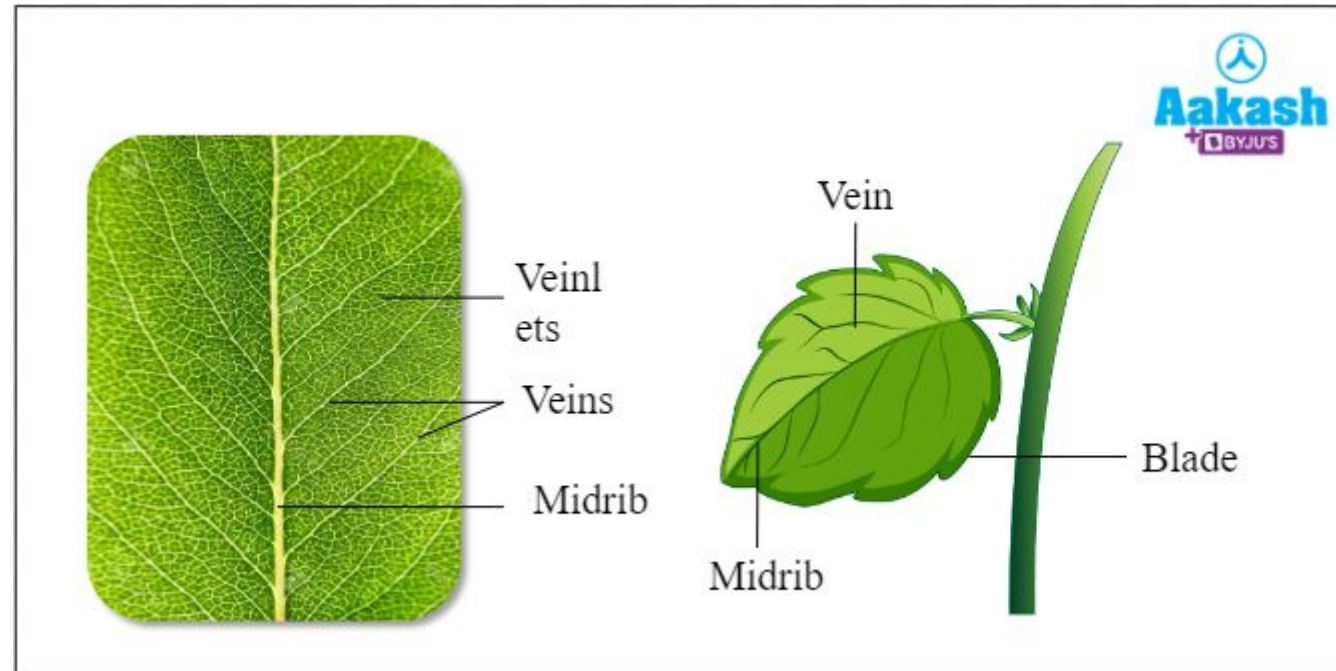
LEAF

- Leaf is a chlorophyll containing flat structure which grows laterally on a stem and develops at the node of a stem. Buds arise from the axil of leaves and later develop into new branches or shoots.
- The leaf is attached to the stem through the leaf base. Stipules are two lateral leaf-like appendages located on either side of the leaf base. Petiole or stalk of the leaf, also known as mesopodium, connects the leaf lamina with the stem or branch. Leaf blade or lamina is the flattened green part of the leaf which is the primary site for photosynthesis.



Structure of the leaf

Through the centre of the leaf runs a prominent vein called midrib. From the midrib arise numerous veins and even smaller veinlets. The midrib, veins and veinlets possess vascular bundles.



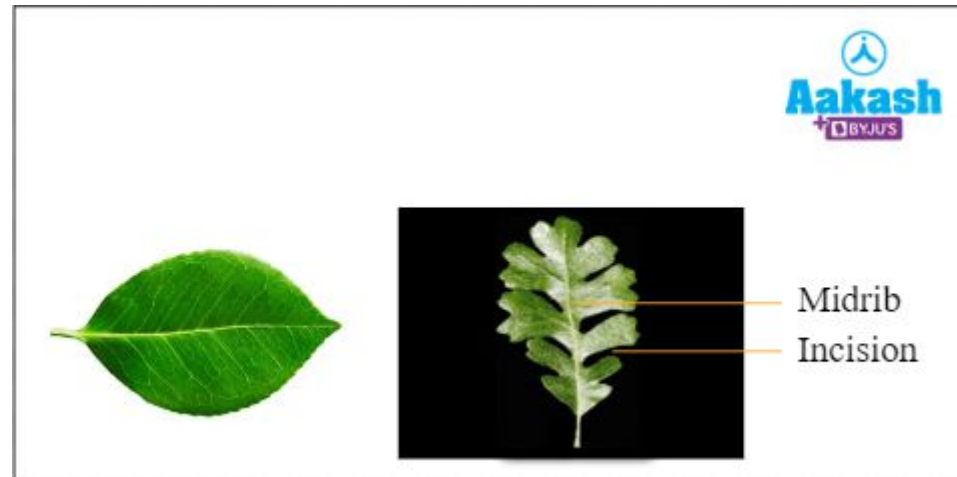
Midrib, veins and veinlets

- **Types of leaves**

- Based on the structure of the leaf blade, leaves are categorised as simple and compound.

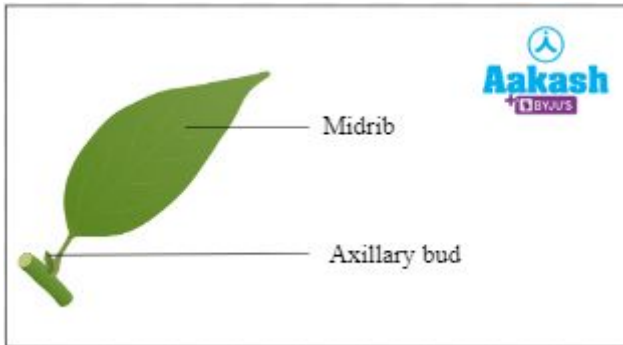
- **Simple leaf**

- In a simple leaf, the leaf blade is not broken into lobes or leaflets as the incisions of the lamina do not touch the midrib.



Simple Leaf

A bud is present in the axil of the petiole in simple leaves, called the axillary or lateral bud.



Simple leaf showing axillary bud

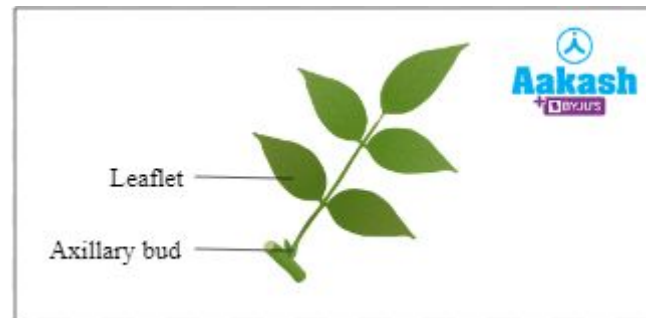
- **Compound leaf**

- In compound leaf, the leaf blade is divided into leaflets as the incisions of the lamina reach up to the midrib, breaking it into a number of leaflets.



Compound Leaf

Axillary bud is present in the axil of the petiole, but not in the axil of leaflets.



Compound leaf showing axillary bud

- Based on leaflet arrangement, compound leaves are further divided as pinnately compound and palmately compound leaves.
- **Pinnately compound leaves**
- In pinnately compound leaves, the midrib forms the common axis known as rachis along which the leaflets grow.. Example: Neem leaves.
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Neem leaves

- Bipinnate - The leaflets are divided to second order and arise on secondary axes on the rachis, e.g, Acacia, Mimosa, etc.
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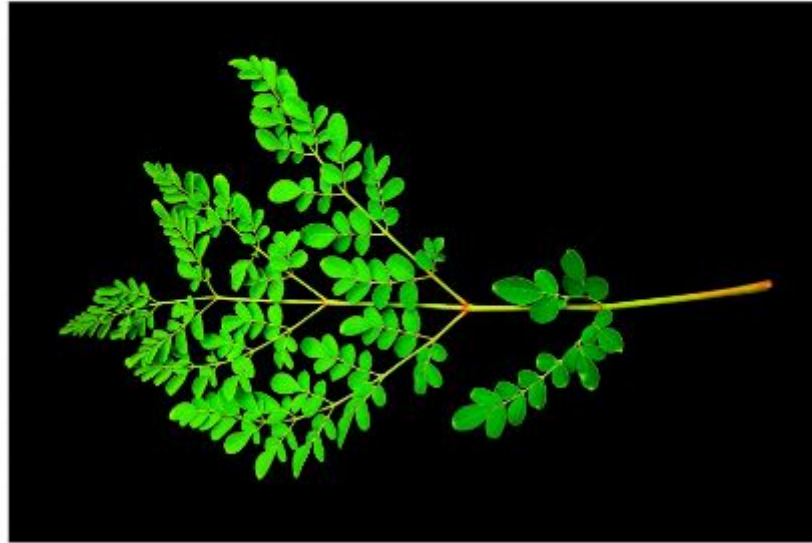


Unipinnate leaf



Bipinnate leaf

- Tripinnate - The leaflets are borne on the tertiary axes which arise from the secondary axes of the rachis, e.g, Moringa.



Tripinnate leaf

Decompound - decompound leaves are more than thrice pinnate, e.g, coriander, carrot, etc.



Decomposed leaf

- **Palmately compound leaves**

- In palmately compound leaves, the leaflets are attached to a common point, i.e., tip of the petiole. The leaflets may be petiolate or sessile. Example: Silk cotton leaves.



Palmately compound Leaf

- Palmately compound leaves can be of the following types -
- Unifoliate - Only a single leaflet is attached to the tip of the petiole, e.g, Citrus.



Unifoliate leaves

- Bifoliate - A pair of leaflets is borne on the rachis, e.g, Balanitis.
-



Bifoliate leaves

- Trifoliate - There are three leaflets borne on the tip of the rachis, e.g, Clover, Oxalis, etc.
-



Trifoliate leaf

- Quadrifoliate - These are four leaflets on the tip of the rachis, e.g, Marsilia.
-



Quadrifoliate leaf

- Multifoliate - The tip of the petiole bears more than four leaflets, e.g, Bombax.
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Multifoliate leaf

- **Phyllotaxy**

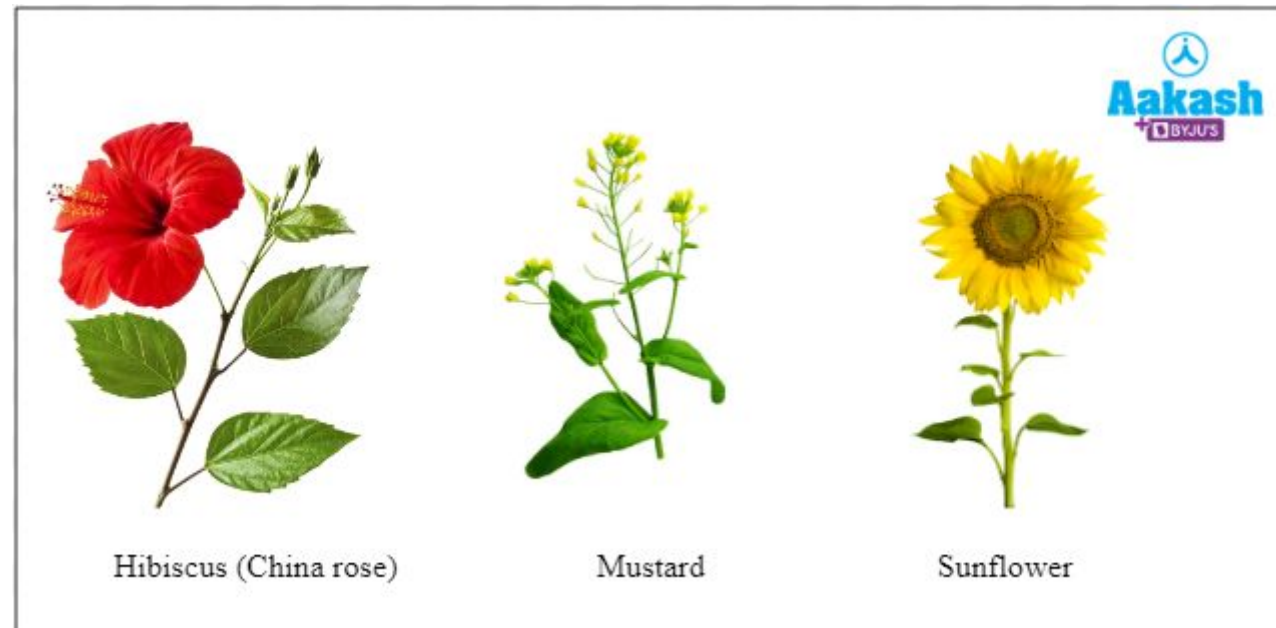
- The pattern of arrangement of leaves on the stem or branch is known as phyllotaxy.

- **Types of phyllotaxy**

- **Alternate**

- In this type of phyllotaxy, only a single leaf arises at each node. Examples: Hibiscus, mustard sunflower etc.

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Alternate phyllotaxy

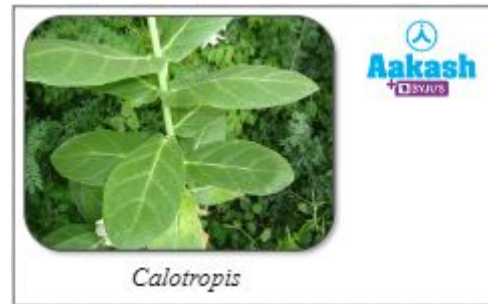
Opposite

In opposite phyllotaxy, each node possesses two leaves lying opposite to each other. Based on the arrangement, opposite phyllotaxy can further be classified as -

Opposite superposed	Opposite decussate
The paired leaves at each node lie exactly above the pair at a lower node.	The paired leaves at one node lie at right angles to the pair above or below them.
Example - guava	Example - Calotropis



Opposite superposed phyllotaxy



Opposite decussate phyllotaxy

- **Whorled**

- In whorled phyllotaxy, more than two leaves arise from a single node. Example: Alstonia

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Whorled phyllotaxy

- **Modifications of leaves**

- Leaves of a plant primarily contribute to preparation of food by photosynthesis but in some special cases, leaves may be modified to form different structures that can help in doing additional functions such as protecting the plant, reducing transpiration, storing food, providing support, trapping and digesting insects for nutrition, etc.

- **Tendrils**

- The leaves are modified as thin, thread-like coiled structures which coil around a support and help in climbing. The tendrils respond to touch and twine around objects.



Tendrils coiling around a support

In garden peas, the terminal leaflets are modified into tendrils.



Pea plant

- **Spines**

- Spines are hard, stiff, and pointed vasculated structures that are modifications of either an entire leaf or leaf parts. In xerophytic plants, the leaves are reduced to sharp pointed structures called spines. This is a modification to conserve water and reduce its loss of water via transpiration by reducing the area of exposed leaf surface. Spines also provide protection against grazing animals. In such plants, the stems are modified into flat green structures which can perform photosynthesis.



Cactus

- **Leaflet hooks**

- The upper three leaflets of Bignonia are modified into curved, stiff, hook-like structures to help the plant adhere to support for climbing.

- **Storage leaves**

- Some leaves are modified to store food. The edible parts of garlic and onion are the fleshy scale leaves that are modified to store food. The outer scale leaves are dry and generally discarded. In succulents such as Aloe, the leaves store mucilaginous substances.

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Leaves modified for storage of food

- **Phyllode (Modified leaf stalk)**

- Phyllodes are modified petioles which are green, flattened and capable of performing photosynthesis. Phyllodes are found in plants such as Australian acacia in which the leaves are small and short-lived.



Australian acacia

- **Carnivorous leaves**

- Leaf lamina of insectivorous plants gets modified to trap insects. These plants are photoautotrophic but grow in soil which lacks nitrogen. Thus, they trap insects and eat them to fulfil their nitrogen requirement. Examples: Venus flytrap (leaf lamina is bilobed and hinged to trap insects), Pitcher plant (leaf lamina is modified as a pitcher).



Nepenthes: Pitcher Plant



Venus fly trap

In Utricularia, some of the leaf segments are modified into small bladders which are specialised to trap small animals such as water fleas.



Bladders of Utricularia

- **Floral leaves and sporophylls**

- The floral parts such as sepals, petals, stamens and carpels are also modified leaves. Stamens and carpels are known as sporophylls as they are modified spore-bearing leaves.